C128 USB-C PD Power Adapter v0.1 (DC only, no 50/60 Hz: the CIA TOD clocks will not tick from mains)
Needs a USB-C PD charger offering 15 V (>= 3 A / 45 W recommended). U1 (CH224A) requests 15 V via R1 = 56k.
Q1 connects the 15 V rail only after the PD contract (PG low). U2 and U3 start only above ~13.3 V (EN UVLO).
DIN (square 5-pin): 1 = +5.10 V / ~3 A (U2 SY8205, D1 TVS), 3 = +12.3 V DC (U3, D2 TVS) into the C128 "9 VAC" bridge,
2 = GND, 4 = GND/NC, 5 = other 9 VAC leg tied to GND (pin numbers per the Drunk'n'Retro diagram).
VERIFY THE SOCKET PINOUT ON YOUR C128 WITH A METER BEFORE FIRST POWER-UP.

PWR_FLAG PWR_FLAG PWR_FLAG PWR_FLAG PWR_FLAG
VBUS GND VSW +5V +12V

VBUS
R2
100k
GATE
GATE
C2
100nF/50V
VSW
Q1
GATE
1
3
2
AO3401A
VBUS
GATE
R3
100k
PG

J1
VBUS
A4
CC1
A5
CC2
B5
D-
A7
D-
B7
D+
A6
D+
B6
SBU1
A8
SBU2
B8
SHIELD
GND
SH
A1
USB-C (power in)
GND GND
VBUS VBUS VBUS VBUS
VBUS
U1
CC1
7
CC2
6
5
DM
4
DP
CC1
VBUS
VDD
PG
CFG1
CFG2
CFG3
8
1
10
9
2
3
11
CH224A
GND
PG
CFG1
VBUS
C1
1uF/50V
GND
CFG1
R1
56k
GND
VSW
C15
10uF/25V
GND
VSW
C3
10uF/25V
GND
VSW
C4
100nF/50V
GND
U2
VSW
7
VSW
8
EN
3
SS
4
IN
IN
EN
SS
GND
BS
LX
FB
VCC
1
2
5
6
9
BOOT
SW
FB
VCC5
SY8205FCC
GND
BOOT
C5
100nF/50V
SW
SW
L1
4.7uH
+5V
+5V
R6
75k
FB
FB
R7
10k
GND
+5V +5V +5V
C6
22uF/25V
GND
C7
22uF/25V
GND
C18
22uF/25V
GND
VCC5
C16
1uF/50V
GND
SS
EN
C17
100nF/50V
GND
R5
51k
GND
12V_BUCK
F2
0R
+12V
D1
+5V
SMF5.0A
GND
D2
+12V
SMF13A
GND
+5V
GND
R8
1k
D3
LED_A
GREEN
LED_A
J2
GND GND
GND
+12V
+5V
1
2
3
4
5
C128 POWER (square DIN-5 male)
VSW
VSW
C9
10uF/25V
GND
C10
100nF/50V
GND
U3
VSW
3
EN2
5
VIN
EN
GND
BOOT
SW
FB
6
2
4
1
BOOT2
SW2
FB2
TPS54302DDCR
GND
12V_BUCK
C14
100pF
FB2
VSW
R9
470k
EN2
R10
47k
GND
BOOT2
C11
100nF/50V
SW2
SW2
L2
10uH
12V_BUCK
12V_BUCK
R11
100k
FB2
FB2
R12
5.1k
GND
12V_BUCK
C12
22uF/25V
GND
12V_BUCK
C13
22uF/25V
GND

USB-C PD 15 V -> C128 square DIN-5: +5 V (SY8205) and +12 V DC (TPS54302) on the 9 VAC inp

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Size: A3	Date: 2026-11-02	**Rev: 0.1**
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